Evidence-Driven Research Poster Rubric (Final Poster Assessment)

HONR 46400 · Purdue Fall Undergraduate Research Expo · grades the locked poster (M13) as the Final Project poster-quality score: 70% of the Poster Presentation component, with the M15 live presentation supplying the other 30%. Built for the course template honr464_poster_template.pptx (34 × 22.5 in). Score each criterion in one band; the hard caps below override the bands. The rubric scores what the poster communicates; check that it is true against the M9 note, the package, the run records and the ledger.

#	Criterion · panels graded	Pts	Exemplary	Proficient	Developing	Beginning	Score
1	**Question, context & contribution** *Title · Abstract · The question and why it matters · What is already known* Compass & pathway alignment · Ch. 1, 7–9	10	**9–10 pts:** One specific research question, and why it matters; the main conclusion of the literature synthesized from at least two sources you read; what your project adds to that conclusion made explicit; the hypothesis you recorded before the analysis ran, with any later analysis labeled exploratory.	**7–8 pts:** The question is clear, but the context or the contribution stays general: one source, or "little is known" without saying what is known.	**4–6 pts:** The question is broad or ambiguous; sources are listed without saying what they establish; no contribution is stated.	**0–3 pts:** A topic without a usable research question.	
2	**Target, design & warranted claims** *Design: what your claim needs* Compass & pathway alignment · Ch. 6–7, 10, 12, 14–19	15	**14–15 pts:** The question's kind (descriptive or causal) and its reach (a named population beyond the data, or the unseen cases a prediction serves) are declared, with prediction named as an objective when it is one; the target quantity is exact; the board states what supports each crossing (a sampling argument for a population, an assignment or identification argument with its assumptions for a cause, a held-out test against a baseline for prediction) and what the evidence cannot yet support, including a causal target that remains unidentified.	**11–13 pts:** Design and target are clear; one warrant or assumption is underexplained.	**6–10 pts:** The link between target, design, and reach is ambiguous, or the kind is mislabeled (for example, a causal question silently relabeled descriptive).	**0–5 pts:** No defensible account of what the design can establish, or a design presented as able to answer a question it cannot.	
3	**Data, provenance & measurement** *Data and measurement* Evidence integrity & provenance · Ch. 13, 20–21	12	**11–12 pts:** Source and route can be identified (archive, URL or DOI, retrieval date, or collection protocol); observations and independent units are distinguished, with any grouping named; the time window is given; each key concept is tied to its measure with the main validity concern; missing data and their handling are reported; the permission or IRB determination is stated.	**9–10 pts:** The evidence is identifiable; one consequential detail is thin.	**5–8 pts:** Important provenance or measurement information is missing.	**0–4 pts:** The evidence cannot be identified, or a material permission problem is unresolved.	
4	**Analysis, checks & verification** *Methods · Checks and verification · verification status line* Verification & robustness · Ch. 11, 22–25, 36–37	15	**14–15 pts:** The analysis fits the design; what was declared in advance is kept apart from later changes, and each change is explained; the uncertainty method fits the data, computed at the level of the independent unit with any dependence named; at least one consequential robustness, sensitivity, or diagnostic check is reported with what it changed; the verification status is accurate and honestly labeled (an external cold run, or "solo proxy; external cold run pending").	**11–13 pts:** Analysis and uncertainty are appropriate; the check or the verification account is thin.	**6–10 pts:** A weak link between method and design, an uninformative check, or a verification claim the record does not support.	**0–5 pts:** A material analytical error, or a displayed result with no trace or that fails to reproduce.	
5	**Results & figure honesty** *Main result* Verification & robustness · Ch. 32	15	**14–15 pts:** The main evidence answers the question and is easy to inspect; key estimates and comparisons are labeled directly, with units and the n; the uncertainty the design warrants is shown (an interval with its type and level, or a precise caveat saying why none is warranted); scales are honest (bars start at zero, other plots carry clearly labeled scales); the caption names the notebook cell or output behind the figure.	**11–13 pts:** The result is understandable and honest; one labeling or caption omission.	**6–10 pts:** The reader has to reconstruct a key comparison, a denominator, or the uncertainty.	**0–5 pts:** The visual encoding distorts the result (a truncated bar axis turning a small gap into a cliff), or no inspectable result is presented.	
6	**Interpretation, uncertainty & limitations** *Headline · Conclusions · Limitations* Uncertainty & limitations · Ch. 28–29, 35	15	**14–15 pts:** Headline and conclusion claim no more than the evidence: causal wording only with a defended identification argument, population wording only with a reach argument; the qualifier that blocks the likely wrong inference sits in the same eye-span as the claim; the consequential limitation and a feasible next step are explicit (uncertainty, limitation, next step); the preliminary status is stated.	**11–13 pts:** The interpretation is sound; the qualifier's placement or the next step is underdeveloped.	**6–10 pts:** Generic limitations ("results may vary"), uncertainty off the read path, or wording that drifts past the evidence.	**0–5 pts:** The central claim exceeds its warrant, or the result is presented as settled certainty.	
7	**Sources, AI disclosure & credit** *AI use disclosure · References · Acknowledgments* Evidence integrity & provenance · Ch. 9, 30	8	**8 pts:** Citations and references correspond and survive checking, and each reference can be found (author, year, title, venue, DOI or URL); the AI disclosure is on the poster itself and names each tool, its purpose, what you changed or kept from its output, and how you checked it (or "No AI used", when true); help, data providers, and funding are credited.	**6–7 pts:** Complete, with one reference or one disclosure line ambiguous.	**3–5 pts:** References or disclosure incomplete enough to impede checking.	**0–2 pts:** A fabricated or materially misattributed source, or a confirmed AI contribution missing from the disclosure.	

#	Criterion · panels graded	Pts	Exemplary	Proficient	Developing	Beginning	Score
8	**Visual communication & accessibility** *The whole board* Craft & communication · Ch. 32–33	10	**9–10 pts:** A clear reading order: the finding is found in seconds, and the question, result, and boundary in a ninety-second read; built on the course template at 34 × 22.5 in, with the guide's minimum type sizes; strong contrast; every distinction carried in two channels (a label or pattern, never colour alone); each figure carries alt text in the file; no template guidance or EXAMPLE material left.	**7–8 pts:** Readable and navigable, with minor craft lapses or one colour-only distinction.	**4–6 pts:** Dense, small, or poorly ordered content that needs the presenter; template text left on the board.	**0–3 pts:** The main message is illegible at print size, or meaning is carried by colour alone throughout.	
	Total	**100**					

Hard caps: a single failure caps the criterion at Beginning, regardless of the rest

- A fabricated source, a source that cannot be identified after reasonable checking, or a materially false attribution to a source → Sources, AI disclosure & credit.
- A displayed result your project computed, with no reproducible trace to a notebook cell or output → Analysis, checks & verification.
- A displayed result or bound that materially fails to reproduce from the submitted package (documented rounding is not a failure) → Analysis, checks & verification.
- An author rerun presented as independent reproduction, or a solo proxy without its label → Analysis, checks & verification.
- A central claim that materially exceeds its identification or reach warrant → Interpretation, uncertainty & limitations.
- A confirmed AI contribution missing from the on-poster disclosure → Sources, AI disclosure & credit.

Apply each cap once, to its named criterion: the score falls to the top of the Beginning band, or stays at the lower score already earned. A temporary link failure is not a fabricated source, and missing ledger rows do not by themselves prove undisclosed AI use. Record the evidence behind any cap in the feedback.

Separate from this score: the poster does not lock without the two blocking gate records (author self-reproduction log and release preflight), and a missing AI Research Ledger is handled in the M13 process score. For an approved group the poster-quality score is common; every member presents and receives their own live score. Source of truth: milestone_13_final_poster_lock.md (generated block).